ModelIt!

Student Activity Pack

Could We Actually Mine Asteroids?

Modeling Space Resource Economics and Orbital Mechanics

Grade 11L1 | NGSS HS-ESS1-4, HS-ETS1-3 | Scale, Proportion, and Quantity

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What do you think asteroids are made of, and could any of those materials be valuable?

2 Why do you think it's so expensive to launch things into space — what makes it harder than traveling on Earth?

3 If an asteroid contained $10 trillion worth of platinum, why hasn't anyone gone to mine it yet?

4 What challenges would a mining operation face in the vacuum of space with no gravity, no air, and extreme temperature swings?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Asteroid Type | Orbital Distance | Mission Cost | Delta-v Requirement | Resource Value | Return on Investment

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Near-Earth Water Mining

C-type asteroid | Delta-v: 4.5 km/s | Resource: water/ice

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Deep Space Platinum

M-type asteroid | Delta-v: 8.0 km/s | Resource: platinum group metals

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Break-Even Analysis

Variable asteroid distances | Fixed resource value

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Market Price Elasticity — How the price of a commodity changes when supply increases dramatically — flooding Earth's market with asteroid platinum would crash the price, potentially making the mission unprofitable even if extraction succeeds

In-Space Fuel Depot — Orbital infrastructure that stores water-derived hydrogen and oxygen fuel for other spacecraft — creating a fueling station in orbit could generate recurring revenue from other space missions, changing the economics from one-time resource extraction to ongoing service provision

Spacecraft Propulsion Type — The engine technology used for the mission — chemical rockets have high thrust but low efficiency, while ion engines have extremely high efficiency but very low thrust, creating trade-offs between trip time and fuel requirements

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

There are asteroids worth more than the entire global economy floating in space — so why haven't we gone to get them, and what would it take? Use evidence from your simulation to support your answers.

1. **How does the Tsiolkovsky rocket equation fundamentally shape the economics of space mining — and why can't you just 'build a bigger rocket' to solve the problem?**

2. **Why might water be more valuable than platinum in the context of space operations — what does this reveal about how value depends on context?**

3. **How does the market crash paradox (bringing rare materials back collapses their price) challenge the traditional mining business model?**

4. **What role does in-space infrastructure play in making asteroid mining viable — and how is this analogous to how railroad infrastructure enabled terrestrial mining?**

5. **What are the ethical and legal implications of private companies claiming ownership of celestial resources?**

8 STEM Challenge: Design a First-Generation Asteroid Mining Mission

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YOUR ENGINEERING MISSION

Design a commercially viable first asteroid mining mission that identifies a target asteroid, calculates delta-v requirements, estimates costs, and proposes what resources to extract and how to use them.

THE SCENARIO

A space mining startup has raised $2 billion in funding and needs to demonstrate a profitable first mission within 7 years. Your team must select a target asteroid from a catalog of near-Earth objects, design the mission profile, calculate the delta-v budget, estimate costs, and prove to investors that the Return on Investment is positive. The key strategic decision: mine platinum for Earth return or mine water for in-space fuel sales?

Engineering Design Thinking

1. WHAT SPECIFIC ASTEROID FROM THE KNOWN NEA CATALOG WOULD YOU TARGET AND WHY?

2. HOW WOULD YOU MINIMIZE MISSION COST WHILE MAXIMIZING RESOURCE EXTRACTION — WHAT ENGINEERING TRADE-OFFS WOULD YOU MAKE?

3. WHAT INFRASTRUCTURE WOULD YOUR FIRST MISSION CREATE THAT REDUCES COSTS FOR SUBSEQUENT MISSIONS?

4. HOW WOULD YOU ADDRESS THE MARKET CRASH PROBLEM IF YOU BRING LARGE QUANTITIES OF RARE MATERIALS BACK TO EARTH?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Aerospace Engineers designing deep-space mission profiles work at NASA, SpaceX, Blue Origin, and emerging space mining companies like AstroForge, earning $90,000-$170,000/year. Planetary Scientists who characterize asteroid compositions using spectroscopic data earn $70,000-$130,000/year. Space Resource Economists are an emerging field analyzing the financial viability of space ventures, earning $80,000-$150,000/year.